CAD DATA ANALYSIS, THEORETICAL BACKGROUND, AND PRE-SIMULATION CALCULATIONS (PRE-ANALYSIS)

Course Project: Numerical Methods (FEM)

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1 CAD Design Data Analysis

1.1 Component 1: Hexagon Head Screw (Hex Bolt)

Based on the engineering drawing `BOLT-M10-030` and the 3D CAD model `bolt.ipt`, the geometric and material specifications of the bolt are defined as follows:

- **Manufacturing Standard:** ISO 4017 – Metric thread $M10 \times 30$ (full thread, pitch $p = 1.5$ mm, tolerance class $6g$, right-hand thread).
- **Hexagon Head Dimensions:** Width across flats (A/F) $s = 16.00$ mm, head height $k = 6.40$ mm.
- **Shank Dimensions:** Under-head length $l = 30.00$ mm, thread end chamfer $1 \times 45°$.
- **Material and Property Class:** Quenched and tempered carbon steel, **Property Class 8.8** per ISO 898-1:
  - Young's modulus: $E_{\text{bolt}} = 210$ GPa $= 210,000$ MPa.
  - Poisson's ratio: $\nu_{\text{bolt}} = 0.30$.
  - Nominal ultimate tensile strength: $\sigma_{u,\text{bolt}} = 800$ MPa.
  - Nominal yield strength: $\sigma_{y,\text{bolt}} = 0.8 \times \sigma_{u,\text{bolt}} = 640$ MPa.

1.2 Component 2: 16 mm Combination Wrench

Based on the engineering drawing `WR-016-200` and the 3D CAD model `wrench_16mm.ipt`, the geometric and material specifications of the wrench are defined as follows:

- **Overall Dimensions and Thickness:** Uniform body thickness $t = 6.50$ mm ($Z \in [0; 6.50]$ mm), overall nominal length $L_{\text{total}} = 199.55$ mm, center-to-center distance $L_{cc} = 169.00$ mm, calculated mass $m = 148.7$ g.
- **Six-Point Ring Head:** Centered at coordinates $(14, 0)$ mm, outer radius $R_{\text{ring}} = 14.00$ mm. Regular six-point hexagon opening with width across flats $A/F = 16.10^{+0.05}_{-0.00}$ mm (nominal clearance with the bolt head $\Delta = 0.10$ mm).
- **Open Jaw Head:** Centered at coordinates $(183, 0)$ mm, outer radius $R_{\text{open}} = 17.00$ mm. Jaw opening 16.10 mm oriented at 15° in the XY plane, with a semicircular root radius $R_{\text{root}} = 8.05$ mm centered at local coordinates $(-3, 0)$ mm.
- **Shank Profile:** The upper boundary connects points $(14, 14) \to (37, 7) \to (160, 7) \to (175, 15)$ mm and is mirrored across the centerline $Y = 0$. The straight shank width is $w = 14.00$ mm, with transition fillet radii $R = 3.00$ mm at $(37, \pm 7)$ mm and $(160, \pm 7)$ mm.
- **Material and Heat Treatment:** Chromium-Vanadium alloy steel **AISI 6150 Cr-V**, quenched and tempered to 44–48 HRC, with a satin nickel/chrome finish:
  - Density: $\rho = 7850$ kg/m^3.
  - Young's modulus: $E_{\text{wrench}} = 210$ GPa $= 210,000$ MPa.
  - Poisson's ratio: $\nu_{\text{wrench}} = 0.30$.
  - Reference yield strength at 44–48 HRC: $\sigma_{y,\text{wrench}} \approx 1300$ MPa.

Table 1: Summary of material properties for FEM simulation

Component	**Material**	E **(MPa)**	ν	σ_y **(MPa)**
Hex Bolt $M10 \times 30$	Carbon Steel Class 8.8 (ISO 898-1)	$210,000$	0.30	640
16 mm Wrench	AISI 6150 Cr-V Steel (44–48 HRC)	$210,000$	0.30	1300

2 Theoretical Background and Pre-Simulation Calculations

2.1 Standard Tightening Torque for $M10 \times 1.5$ Bolt

According to ISO metric thread standards, the pitch diameter d_2 and minor diameter d_3 of an $M10 \times 1.5$ thread ($d = 10$ mm, $p = 1.5$ mm) are given by:

$$d_2 = d - 0.6495p = 10 - 0.6495 \times 1.5 = 9.026 \text{ mm} \tag{1}$$

$$d_3 = d - 1.2268p = 10 - 1.2268 \times 1.5 = 8.160 \text{ mm} \tag{2}$$

The nominal tensile stress area A_s of the threaded section is:

$$A_s = \frac{\pi}{4}\left(\frac{d_2 + d_3}{2}\right)^2 = \frac{\pi}{4}\left(\frac{9.026 + 8.160}{2}\right)^2 \approx 58.0 \text{ mm}^2 \tag{3}$$

When tightening a Property Class 8.8 bolt ($\sigma_{y,\text{bolt}} = 640$ MPa) to 70% of its yield strength in accordance with standard mechanical assembly guidelines, the assembly preload force F_M is:

$$F_M = 0.7 \cdot \sigma_{y,\text{bolt}} \cdot A_s = 0.7 \times 640 \times 58.0 = 25,984 \text{ N} \approx 26.0 \text{ kN} \tag{4}$$

Assuming a nut factor (torque coefficient) $K = 0.19$ for standard steel-on-steel fastener surfaces, the nominal tightening torque T_{nom} required at the bolt head is:

$$T_{\text{nom}} = K \cdot d \cdot F_M = 0.19 \times 0.010 \text{ m} \times 25,984 \text{ N} \approx 49.37 \text{ N} \cdot \text{m} \approx 50 \text{ N} \cdot \text{m} \tag{5}$$

2.2 Formulation of Applied Load Cases

Assuming the operator engages the six-point ring head of the wrench (centered at $x_{\text{ring}} = 14$ mm) onto the bolt head and applies a perpendicular hand force at the open-jaw end (centered at $x_{\text{open}} = 183$ mm), the effective moment arm is:

$$L_{cc} = x_{\text{open}} - x_{\text{ring}} = 183 - 14 = 169 \text{ mm} = 0.169 \text{ m} \tag{6}$$

The relationship between the applied hand force F_{hand} and the resulting torque T is:

$$F_{\text{hand}} = \frac{T}{L_{cc}} \tag{7}$$

To comprehensively evaluate the structural performance across operating regimes, three load cases are established as summarized in Table 2.

Table 2: Load cases analyzed in the FEM simulation

Load Case	**Operating Condition**	**Torque T ($\mathbf{N \cdot m}$)**	**Torque T ($\mathbf{N \cdot mm}$)**
Case 1	Light tightening (60% rated torque)	30.0	30,000
Case 2	Standard Class 8.8 tightening	50.0	50,000
Case 3	Peak overload / Seized bolt breakaway	80.0	80,000

2.3 Analytical Verification via Mechanics of Materials

2.3.1 Nominal Bending Stress in the Wrench Shank

Consider the critical cross-section at the transition between the wrench shank and the ring head ($x = 37$ mm, where the shank width is $w = 14.0$ mm, thickness $t = 6.5$ mm, and fillet radius $R = 3.0$ mm). The moment arm from the applied hand load ($x = 183$ mm) to this cross-section is:

$$L_b = 183 - 37 = 146 \text{ mm} \tag{8}$$

The area moment of inertia I_z and elastic section modulus W_z of the rectangular cross-section $t \times w$ are:

$$I_z = \frac{t \cdot w^3}{12} = \frac{6.5 \times 14.0^3}{12} \approx 1486.33 \text{ mm}^4 \tag{9}$$

$$W_z = \frac{t \cdot w^2}{6} = \frac{6.5 \times 14.0^2}{6} \approx 212.33 \text{ mm}^3 \tag{10}$$

Under the standard load case $T_2 = 50$ N $\cdot$ m ($F_2 = 295.9$ N), the bending moment at $x = 37$ mm and the maximum nominal bending stress at the outer fibers ($y = \pm 7$ mm) are:

$$M_b = F_2 \cdot L_b = 295.9 \times 146 = 43,201.4 \text{ N} \cdot \text{mm} \tag{11}$$

$$\sigma_{\text{nom}} = \frac{M_b}{W_z} = \frac{43,201.4}{212.33} \approx 203.46 \text{ MPa} \tag{12}$$

Accounting for geometric stress concentration at the transition fillet $R = 3$ mm ($R/w = 3/14 \approx 0.214$), the theoretical stress concentration factor is $K_t \approx 1.45$–1.60. Thus, the expected peak local stress at the wrench neck is:

$$\sigma_{\text{max,theory}} = K_t \cdot \sigma_{\text{nom}} \approx (1.45 \div 1.60) \times 203.46 \approx 295 \div 325 \text{ MPa} \tag{13}$$

Approximating the shank as a cantilever beam of length $L_b = 146$ mm, the tip deflection contribution of the shank is:

$$u_{y,\text{shank}} = \frac{F_2 \cdot L_b^3}{3E_{\text{wrench}} I_z} = \frac{295.9 \times 146^3}{3 \times 210,000 \times 1486.33} \approx 0.984 \text{ mm} \tag{14}$$

2.3.2 Torsional Shear Stress in the Bolt Shank

Under pure torsion $T_2 = 50,000$ N $\cdot$ mm acting on the equivalent stress diameter of the bolt shank $d_s = (d_2 + d_3)/2 \approx 8.60$ mm, the polar section modulus W_p and maximum torsional shear stress $\tau_{\max}$ are:

$$W_p = \frac{\pi d_s^3}{16} = \frac{\pi \times 8.60^3}{16} \approx 124.89 \text{ mm}^3 \tag{15}$$

$$\tau_{\max} = \frac{T_2}{W_p} = \frac{50,000}{124.89} \approx 400.35 \text{ MPa} \tag{16}$$

The corresponding theoretical Von Mises stress on the outer surface of the bolt shank under pure torsion is:

$$\sigma_{\text{vM, shank}} = \sqrt{3}\,\tau_{\max} \approx 693.4 \text{ MPa} \tag{17}$$

2.4 Solid Mechanics Governing Equations and Finite Element Formulation (FEniCS)

2.4.1 Governing Differential Equations (Strong Form)

Consider a linear elastic, isotropic solid occupying a spatial domain $\Omega \subset \mathbb{R}^3$ with boundary $\partial\Omega = \Gamma_D \cup \Gamma_N$. Neglecting body forces ($\mathbf{f} = \mathbf{0}$), the static equilibrium equation is given by:

$$-\nabla \cdot \boldsymbol{\sigma}(\mathbf{u}) = \mathbf{0} \quad \text{in } \Omega \tag{18}$$

The infinitesimal Cauchy strain tensor $\boldsymbol{\varepsilon}(\mathbf{u})$ is related to the displacement vector $\mathbf{u} = (u_x, u_y, u_z)^T$ by:

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right) \tag{19}$$

The constitutive equation (generalized Hooke's law) for a linear elastic isotropic material is:

$$\boldsymbol{\sigma}(\mathbf{u}) = \lambda\, \mathrm{tr}(\boldsymbol{\varepsilon}(\mathbf{u}))\mathbf{I} + 2\mu\, \boldsymbol{\varepsilon}(\mathbf{u}) \tag{20}$$

where $\mathbf{I}$ is the second-order identity tensor, and λ, μ are the Lamé parameters expressed in terms of Young's modulus E and Poisson's ratio ν:

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = G = \frac{E}{2(1+\nu)} \tag{21}$$

For $E = 210,000$ MPa and $\nu = 0.30$:

$$\lambda = 121,153.85 \text{ MPa}, \qquad \mu = 80,769.23 \text{ MPa} \tag{22}$$

The boundary conditions on $\partial\Omega$ are prescribed as:

$$\mathbf{u} = \mathbf{0} \qquad \text{on the Dirichlet boundary } \Gamma_D \text{ (fixed/constrained contact surfaces)} \tag{23}$$

$$\boldsymbol{\sigma}(\mathbf{u}) \cdot \mathbf{n} = \mathbf{t} \qquad \text{on the Neumann boundary } \Gamma_N \text{ (applied surface traction)} \tag{24}$$

2.4.2 Variational Formulation (Weak Form) in FEniCSx

Let $V = \{\mathbf{v} \in [H^1(\Omega)]^3 \mid \mathbf{v}|_{\Gamma_D} = \mathbf{0}\}$ denote the trial and test function space discretized using quadratic Lagrange tetrahedral elements ($\mathcal{P}_2$ – equivalent to 10-node `C3D10` elements). Multiplying the equilibrium equation by a test function $\mathbf{v} \in V$ and integrating by parts via the Gauss–Green divergence theorem over Ω yields the variational problem:

Find $\mathbf{u} \in V$ such that $\forall \mathbf{v} \in V$:

$$a(\mathbf{u}, \mathbf{v}) = L(\mathbf{v}) \tag{25}$$

where the bilinear form $a(\mathbf{u}, \mathbf{v})$ and linear form $L(\mathbf{v})$ are defined as:

$$a(\mathbf{u}, \mathbf{v}) = \int_\Omega \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) \, \mathrm{d}\Omega \tag{26}$$

$$L(\mathbf{v}) = \int_{\Gamma_N} \mathbf{t} \cdot \mathbf{v} \, \mathrm{d}\Gamma \tag{27}$$

2.4.3 Von Mises Yield Criterion and Safety Factor

From the computed displacement field $\mathbf{u}_h$, the deviatoric stress tensor $\mathbf{s}$ is obtained by subtracting the hydrostatic stress component:

$$\mathbf{s} = \boldsymbol{\sigma}(\mathbf{u}_h) - \frac{1}{3}\,\mathrm{tr}(\boldsymbol{\sigma}(\mathbf{u}_h))\mathbf{I} \tag{28}$$

The equivalent Von Mises stress at each integration point is evaluated based on the maximum distortion energy criterion:

$$\sigma_{\mathrm{vM}} = \sqrt{\frac{3}{2}\,\mathbf{s} : \mathbf{s}} = \sqrt{\frac{1}{2}\left[(\sigma_{xx} - \sigma_{yy})^2 + (\sigma_{yy} - \sigma_{zz})^2 + (\sigma_{zz} - \sigma_{xx})^2 + 6(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2)\right]} \tag{29}$$

The structural Safety Factor (SF) relative to the material yield strength σ_y is evaluated as:

$$SF = \frac{\sigma_y}{\sigma_{\mathrm{vM,\,max}}} \geq [SF]_{\mathrm{allow}} \tag{30}$$